

Practice Problems, Exam II Solutions

STA-360/602, Spring 2018

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1 Solutions

1. (15 points)

3.14, part d. (Unit information prior).

Similarly to how we write the likelihood up to proportionality, I will define the log-likelihood up to an additive constant that doesn't contain the parameter.

a)

$$\begin{aligned} p(y_1, \dots, y_n) &\propto \prod_{i=1}^n \theta^{y_i} e^{-\theta} \\ &= \theta^{n \bar{y}} e^{-n \theta} \\ l(\theta|y) &= n \bar{y} \log \theta - n \theta \\ \frac{d}{d\theta} l(\theta|y) &= \frac{n \bar{y}}{\theta} - n \\ \frac{d^2}{d\theta^2} l(\theta|y) &= -\frac{n \bar{y}}{\theta^2} < 0 \end{aligned}$$

Setting the derivative of the log-likelihood equal to zero gives us that $\hat{\theta} = \bar{y}$.

$$\begin{aligned} J(\theta) &= -\frac{d}{d\theta} \left(\frac{n \bar{y}}{\theta} - n \right) \\ &= \frac{n \bar{y}}{\theta^2} \end{aligned}$$

so $J(\hat{\theta}) = \frac{n}{\bar{y}}$.

b)

$$\begin{aligned} \log p_U(\theta) &= \frac{l(\theta|y)}{n} + c \\ &= \frac{n \bar{y} \log \theta - n \theta}{n} + c \\ &= \bar{y} \log \theta - \theta + c \end{aligned}$$

which implies that

$$p_U(\theta) \propto \theta^{\bar{y}} e^{-\theta}$$

which implies that p_U is $\text{Gamma}(\theta; \bar{y} + 1, 1)$.

We then get that

$$\begin{aligned} -\frac{\partial^2}{\partial \theta^2} \log p_U(\theta) &= -\frac{\partial^2}{\partial \theta^2} (\bar{y} \log \theta - \theta + c) \\ &= -\frac{\partial}{\partial \theta} \left(\frac{\bar{y}}{\theta} - 1 \right) \\ &= \frac{\bar{y}}{\theta^2} \end{aligned}$$

Notice that this has $\frac{1}{n}$ times the information in the likelihood. In other words, if we think of the likelihood as having n units of information (1 for each observation), then p_U has 1 unit of information. Also note that we arrived at p_U by raising the likelihood to the power $\frac{1}{n}$.

c) I'll use notation as though it is a posterior just for convenience.

$$\begin{aligned} p(\theta | y_1, \dots, y_n) &\propto p(y_1, \dots, y_n | \theta) p_U(\theta) \\ &\propto \theta^n \bar{y} e^{-n\theta} \theta^{\bar{y}} e^{-\theta} \\ &\propto \theta^{(n+1)\bar{y}} e^{-(n+1)\theta} \end{aligned}$$

So $\theta | y_1, \dots, y_n \sim \text{Gamma}((n+1)\bar{y} + 1, n+1)$. One could argue that we shouldn't call this a posterior distribution because the construction of the prior involved the observed data and thus isn't technically a prior.

2. (15 points) (Normal-Normal) Derive the posterior predictive density $p(x_{n+1} | x_{1:n})$ for the Normal-Normal model covered in lecture. Hint: There is an easy way to do this and a hard way. To make the problem easier, consider writing $X_{n+1} = \theta + Z$ given $x_{1:n}$, where $Z \sim \mathcal{N}(0, \lambda^{-1})$.

$$\begin{aligned} E(X_{n+1} | X_{1:n}, \lambda^{-1}) &= E(\theta | X_{1:n}, \lambda^{-1}) + E(Z | X_{1:n}, \lambda^{-1}) = M \\ V(X_{n+1} | X_{1:n}, \lambda^{-1}) &= V(\theta | X_{1:n}, \lambda^{-1}) + V(Z | X_{1:n}, \lambda^{-1}) = L^{-1} + \lambda^{-1} \\ X_{n+1} | X_{1:n}, \lambda^{-1} &\sim N(M, L^{-1} + \lambda^{-1}) \end{aligned}$$

3. For labs 4 – 6, see the solutions.

4. Approach 1 (Simple, but not great)

To draw a sample Z from the distribution of $X | X < c$,

- (a) sample $U \sim \text{Uniform}(0, 1)$,
- (b) set $X = F^{-1}(U)$,
- (c) if $X \geq c$ then return to step 1 (reject), otherwise, output $Z = X$ as a sample (accept).

Why does it work? By the inverse c.d.f. method, we know

$$X = F^{-1}(U) \sim \mathcal{N}(0, 1).$$

By the rejection principle, if we reject any samples X such that

$$X \geq c,$$

then what remains has the conditional distribution given

$$X < c.$$

This approach is not ideal, however, since the rejection rate may be very high, especially when $c \ll 0$.

Approach 2 (Better)

To draw a sample Z from the distribution of $X \mid X < c$,

- (a) sample $U \sim \text{Uniform}(0, 1)$,
- (b) set $V = F(c)U$, and
- (c) set $Z = F^{-1}(V)$.

Why does this work? Note that in Approach 1, rejecting when

$$X \geq c$$

is identical to rejecting when

$$U \geq F(c),$$

and by the rejection principle, we know that the distribution of the U 's that remain after rejection is

$$U \mid U < F(c),$$

in other words, $\text{Uniform}(0, F(c))$.

But that means that the rejection step can be by-passed completely by just sampling

$$V \sim \text{Uniform}(0, F(c))$$

and setting

$$Z = F^{-1}(V).$$

Thus, we can directly sample

$$V \sim \text{Uniform}(0, F(c)),$$

by drawing

$$U \sim \text{Uniform}(0, 1)$$

and setting

$$V = F(c)U.$$

5. Prove that the full conditional distributions $f(x \mid y)$ and $f(y \mid x)$ determine the joint distribution of $f(x, y)$. Specifically, show that the joint can be written as

$$f(x, y) = \frac{f(y \mid x)}{\int [f(y \mid x) / f(x \mid y)] dy}.$$

Note that the joint distribution can be written as follows:

$$f(x, y) = f(y \mid x)f(x) \tag{1.1}$$

$$= f(x \mid y)f(y) \tag{1.2}$$

Consider

$$\text{Denominator} = \int \frac{f(y \mid x)}{f(x \mid y)} dy \tag{1.3}$$

$$= \int \frac{f(x, y)}{f(x)} / \frac{f(x, y)}{f(y)} dy \tag{1.4}$$

$$= \int \frac{f(y)}{f(x)} dy \tag{1.5}$$

$$= \frac{1}{f(x)} \tag{1.6}$$

This completes the argument.

Reference: This proof can be found in Casella and Robert, Monte Carlo Statistical Methods. The proof is known as the Hammersley-Clifford Theorem, which is a result when the conditional distributions determine the joint. (This occurs for multi-Gibbs stage models, so this proof can be extended to the multi-stage case).